INCH-POUND
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SUPERSEDING
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DETAIL SPECIFICATION

HEADSET-MICROPHONE KIT, MK-1564( )/AIC

Inactive for new design after January 9, 1991

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Headset-Microphone Kit MK-1564( )/AIC. The kit is designed to fit both the regular and extra-large size Helmet, Flying Type SPH-4.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-228	-	Cable and Wire, Insulated; Methods of Testing
FED-STD-595/27038	-	Miscellaneous, Semigloss

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-C-572	-	Cords, Yarns and Monofilaments Organic Synthetic Fiber.
MIL-DTL-14072	-	Finishes for Ground Based Electronic Equipment.
MIL-DTL-27072/9	-	Cable, Special Purpose, Electrical, Multiconductor (4 Conductors) and Single Shielded, M27072/9HBKB*
MIL-M-49199/3	-	Microphones, Linear M49199/3
MIL-DTL-55668	-	Cord, Electrical; Audio, Subminiature (Retractile and Straight)

Since Navy-EC is the Navy custodian for this document, all Navy review activities should forward their comments directly to DLA Land and Maritime. Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data that may improve this document should be sent to: DLA-CC, DLA Land and Maritime, ATTN: VAI, P.O. Box 3990, Columbus, Ohio 43218-3990 or emailed to sound@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-202	-	Electronic and Electrical Component Parts
MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-810	-	Environmental Test Methods for Aerospace and Ground Equipment
MIL-STD-1285	-	Marking of Electrical and Electronic Parts
MIL-STD-13231	-	Marking of Electronic Items.

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-454	-	General Guidelines for Electronic Equipment.
MIL-HDBK-781	-	Reliability Test Methods, Plans, and Environments for Engineering Development, Qualification, and Production
MIL-HDBK-831	-	Preparation of Test Reports
MIL-HDBK-5400	-	Electronic Equipment, Airborne, General Guidelines for

(Copies of these documents are available online at http://quicksearch.dla.mil or from the Document Automation and Production Service (DAPS) Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

ARMY

DLD6001601	-	Headset-Microphone Kit, MK-1564( )/AIC ASSEMBLY. (CAGE 57045)
TOP1-2-612	-	Nuclear Environmental Survivability (CAGE 3VJM4)

NAVY

SK-N-864	-	Simulated Gun Blast Producing Equipment. (CAGE 80063)

(Copies of these documents required by contractors in connection with specific acquisition functions may be obtained from the procuring activity at DSCC.cddwgs@dla.mil, or as directed by the contracting officer.)

(Copies of specifications, standards, drawings, handbooks, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI S3.2	-	Speech over Communication Systems, Method for Measuring the Intelligibility of
ANSI S12.6	-	Measurement of the Real-Ear Attenuation of Hearing Protectors at Hearing Protectors

(Copies of these documents are available online at http://www.ansi.org or from the American National Standard Institute, 25 West 43 Street, 4th Floor, New York, NY 10036.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 10012	-	Measurement management systems - Requirements for measurement processes and measuring equipment

(Copies of these documents are available online at http://www.ansi.org or from the American National Standard Institute, 25 West 43 Street, 4th Floor, New York, NY 10036.)

NCSL INTERNATIONAL

NCSL-Z540.3	-	Requirements for the Calibration of Measuring and Test Equipment

(Copies of this document are available online at http://www.ncsli.org or from NCSL International, 2995 Wilderness Place, Suite 107 Boulder, Colorado 80301-5404.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements of MIL-HDBK-5400, for class 1 (50,000 feet and sea level) equipment, should apply as guidelines of this specification, with the exceptions and additions called out herein. When the two documents conflict, this specification shall govern.

3.1.1 Approval of nonstandard parts and materials. The guidelines of MIL-HDBK-5400 for category III production equipment should apply.

3.2 First article (see 4.5). When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval.

3.3 Standard model. When stated in the contract or purchase order, a model equipment, MK-1564( ) /AIC, will be made available to the contractor on loan to be used as a general guide except in instances where specific improvements are detailed herein or where requirements dictate changes. The model equipment shall not be subjected to any destructive tests nor shall any part or component (with the exception of subassemblies retained by machine screws, or the like) be removed or disassembled (see table I).

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TABLE I. Standard model.

Item	Description	Quantity
1	Microphone, Linear M-162/AIC	2
2	Helmet, Flyers, SPH-4	2
3	Earcup Transducer Assembly, Linear	4
4	Helmet, Flyers, SPH-4 1/	10

1/ (Without communications kit)

3.4 Component parts. The MK-1564( )/AIC shall be supplied with the components listed in table II.

TABLE II. Component parts.

Quantity	Description
1	Microphone, Linear M-162/AIC, Boom Assembly (Standard Ball and Socket Type), and Cord Assembly Electrical 89282 (14 Inches In length) in accordance with MIL-M-49199/3.
2	Linear earcup transducer assembly MK-18199( )/AIC, including Earphone Element H-347( )/AIC, in accordance with this specification.
1	Cord Assembly Kit, MK-1564( )/AIC in accordance with this specification (see figure 1 and figure 2).

3.5 Design. The MK-1564( )/AIC shall be stable in mechanical construction, electrical characteristics, and acoustical performance.

3.5.1 Aging. The MK-1564( )/AIC shall not contain parts fabricated from materials known to change their properties appreciably because of aging.

3.5.2 Nuclear survivability. Nuclear survivability shall be considered in the selection of materials to be utilized in the fabrication of MK-1564( )/AIC. The units shall be designed toward withstanding the nuclear environment specified in appendix B to TOP1-2-612.

3.5.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.5.4 Pure tin. The use of pure tin, as an underplate or final finish, is prohibited both internally and externally. Tin content of handset components and solder shall not exceed 97 percent, by mass. Tin shall be alloyed with a minimum of 3 percent lead, by mass (see 6.6).

3.5.5 Headset-microphone assembly. The headset-microphone assembly shall meet the requirements shown on figure 1 and in U.S. Army drawing list DLD6001601.

3.6 Construction. The MK-1564( )/AIC shall be assembled as configured on figure 1.

3.6.1 Reduction on material. Natural rubber or natural rubber cement shall not be used in the MK-1564( )/AIC. Rubber substitutes and plastic materials shall not be used as wedges and fillers.

3.6.2 Earphone diaphragm material. The earphone diaphragm material shall be plastic (Mylar Type A) or equivalent material giving demonstrated similar results, as specified in 3.7.2.1.

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3.6.3 Earphone voice coil material. The earphone voice coil shall be constructed of copper wire. The terminals shall be made of brass.

3.6.4 Earphone cushion and filler material. The cushion material shall be a non-hardening type with a slow memory which takes a temporary “set”. The cushion cover shall be polyurethane.

3.6.5 Earcup material. The earcup shall be constructed of ABS and polycarbonate materials.

3.6.6 Cement. To insure uniform adhesion and strength characteristics, any cement used shall be properly and uniformly controlled and cured throughout production.

3.6.7 Cord

.

3.6.7.1 Conductors. The nominal area of each of the four conductors shall be AWG No. 25. Each conductor shall consist of bunch-stranded AWG No. 40 (0.003145 inch diameter) tinned cadmium-bronze wire. See 3.5.4. The strands shall be twisted to have a right hand lay of approximately one-half inch. The individual strands, before bunching and plating, shall have a minimum tensile strength of 90,000 pounds per inch squared (lbs./in^2) and have a conductivity grade of at least 80 percent. The two conductors of each pair shall be twisted together to have a length of lay of 3/8 inch $\pm$ 1/4 inch. Each pair of conductors shall be separated by 1 inch from the dual helmet connector to the separation point of the earphone and microphone pairs. See figure 1 and figure 2 for assembly and schematic representations.

3.6.7.2 Insulation. The insulation shall consist of extruded Teflon not less than 0.006 inch thick. The insulation shall withstand a potential of 1,000 volts of a commercial line frequency, when applied in accordance with method 6211 of FED-STD-228.

3.6.7.3 Shield. A shield of closely woven braid of nickel plated copper strands shall be applied to provide coverage of not less than 90 percent. See MIL-DTL-27072/9, for shield requirement.) The strands shall conform to the requirements for conductors (see 3.6.7.1). The metallic coating of the shield shall be similar to the metallic coating of the conductors of the wire, to which the shield is applied. The braided shield shall not increase the maximum diameters of the type specified, by more than 0.030 inch. The shield shall provide both electrostatic and electromagnetic protection for the internal conductors. At the point where the four (4) conductor cable separates for the earphones and microphone, the shield going toward earphones shall terminate. See figure 1, for a schematic of the MK-1564( )/AIC cabling.

3.6.7.4 Color code. The insulation for the four conductors shall consist of 1 green and 1 red for microphone high and low and 1 white and 1 black for headset high and low, respectively.

3.6.7.5 Staycord. The staycord shall consist of a fiber conforming to MIL-C-572, type PAA. The cord shall have a minimum breaking strength of 20 pounds for cords of 2, 3, and 4 conductors. The filaments of the staycord shall be bonded together with, a fungus-inert adhesive in order to facilitate knotting during termination of the finished cord.

3.6.7.6 Jacket. The cable assembly shall be covered with a vulcanized rubber jacket in accordance with MIL-DTL-55668.

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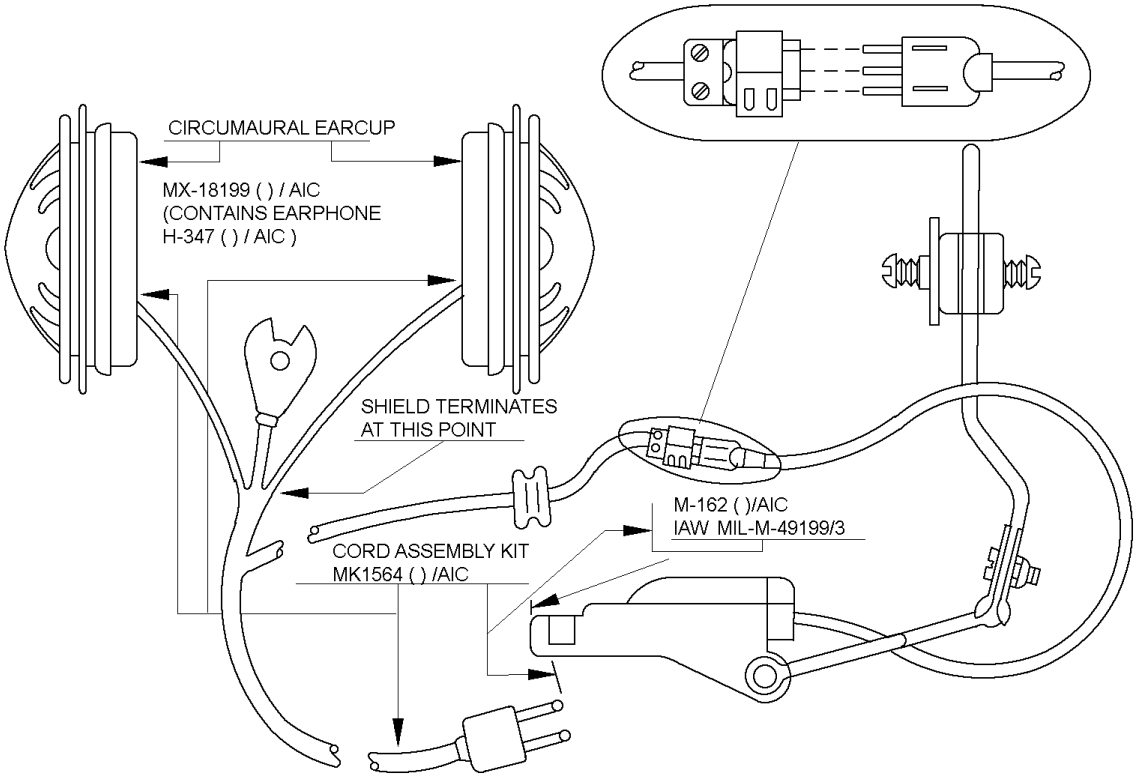


FIGURE 1. MK-1564( )/AIC, assembly of.

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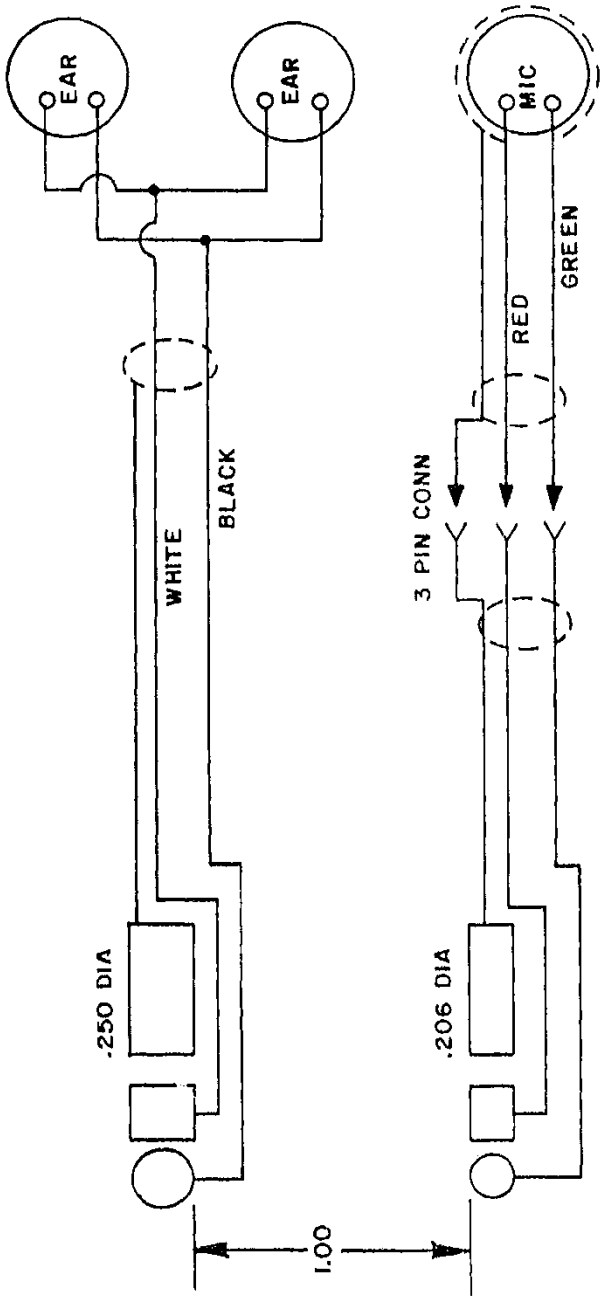


FIGURE 2. Schematic of helmet plug and MK-1564( )/AIC wiring with ferromagnetic shielded cable.

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3.7 Performance.

3.7.1 Electrical performance of earcup transducer assembly.

3.7.1.1 Impedance (see 4.7.2.1). The impedance of the assembly at 1 kHz shall be 19 ohms ± 10 percent.

3.7.1.2 Insulation resistance (see 4.7.2.2). The Insulation resistance of the assembly between the terminals and the metal cover shall not be less than 10 megohms.

3.7.1.3 Dielectric withstanding voltage (see 4.7.2.3). The assembly shall withstand, without flashover or breakdown, the application of a 100 volts alternating potential of commercial line frequency, when tested as specified herein.

3.7.2 Acoustical performance of earcup transducer assembly.

3.7.2.1 Frequency response (see 4.7.3.1 and 4.4.1). The frequency response of the earphone, when installed in its earcup and with earphone cushion in place shall be 85 dB SPL with 1 mW at 1 kHz and within ± 5 dB between 300-6,000 Hz. NOTE: Real head frequency response on a typical person shall not vary from the limits described herein.

3.7.2.2 Psychoacoustic (see 4.7.3.2). The MK-1564( )/AIC shall enable the listeners to achieve a score of 90 percent or better.

3.7.2.3 Harmonic distortion (see 4.7.3.3). Harmonic distortion measurements shall be made on the earphone at 85, 95 and 105 dB SPL. At no level shall 5 percent harmonic distortion be exceeded.

3.7.2.4 Attenuation capability of earcup transducer assembly.

3.7.2.4.1 Real head attenuation (see 4.7.3.4.1). The attenuation values for the earcup transducer assembly shall be equal to or greater than the values shown in the “Real head” column of table III.

3.7.2.4.2 Flat plate coupler attenuation (see 4.7.3.4.2). The attenuation values for the earcup transducer assembly shall be equal to or greater than the values shown in the “Flat plate coupler” column of table III.

3.7.2.5 Linearity (see 4.7.3.5). The response output of the assembly shall be a linear function (± 1 dB) of input, as the input voltage is varied for acoustic outputs of 85, 95, and 105 dB SPL at 1 kHz.

3.7.3 Cord, cordage, and cables.

3.7.3.1 Temperature range (see 4.7.4.1). The unterminated cable shall be flexible and resilient throughout the temperature range of - 55 degrees C to + 85 degrees C and shall show no evidence of cracking or any other damage as a result of exposure to the temperature range specified herein.

3.7.3.2 Flexing life (see 4.7.4.2). The unterminated cable shall have a mean flex life of 300,000 flexes without showing evidence of:

a. Damage to the outside jacket.
b. Damage to the insulation of the individual conductors.
c. Electrical discontinuity.

3.7.3.3 Cord anchorage (see 4.7.4.3). The cord and its anchorage to the two connector plugs shall withstand a specified pull applied to each connector plug for a period of 30 minutes without damage to the

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cord or anchorage. The specified weights shall be an 8-pound pull applied to the microphone connector and a 20-pound pull applied to the main cable kit terminating connector."

3.7.3.4 Conductor strain relief and twist relief (see 4.7.4.4). The cord shall be anchored in the two connector plugs, so as to provide strain relief for the conductors. Means shall be provided to prevent twisting or inward movement of the cord in the connector plug.

TABLE III. Attenuation values.

Frequency (Hz)	Attenuation (dB SPL)	
	Real head 1/	Flat plate coupler
100	-7	
125	-6	
160	-7	
200	-15	-20
250	-18	-18
315	-23	-26
400	-27	-27
500	-30	-34
630	-34	-35
800	-39	-35
1,000	-36	-35
1,250	-35	-48
1,600	-40	-36
2,000	-44	-37
2,500	-40	-38
3,150	-44	-41
4,000	-46	-45
5,000	-48	-38
6,300	-47	
8,000	-44	
10,000	-43	

1/ A +5 dB variation from these values shall be acceptable, provided the values in the "flat plate coupler" column are met or exceeded.

3.7.3.5 Dielectric strength of cord assembly (see 4.7.4.5). For each length of finished cord assembly, the insulation between conductors shall withstand, without flashover or breakdown, the application of a 500 volt alternating potential of commercial line frequency for 5 seconds.

3.7.3.6 Insulation resistance of cord assembly (see 4.7.4.6). The Insulation resistance of the finished cord assembly shall be in accordance with MIL-DTL-55668,

3.7.4 Environmental.

3.7.4.1 Temperature altitude (see 4.7.5.1). The MK-1564( )/AIC shall meet the requirements of 3.7.2.1.

3.7.4.2 Humidity (see 4.7.5.2). The MK-1564( )/AIC shall meet the requirements of 3.7.2.1.

3.7.4.3 Immersion (see 4.7.5.3). The MK-1564( )/AIC shall meet the requirements of 3.7.2.1.

3.7.4.4 Dust (see 4.7.5.4). The MK-1564( )/AIC shall meet the requirements of 3.7.2.1.

3.7.4.5 Salt fog (see 4.7.5.5). The MK-1564( )/AIC shall show no corrosion of finishes and metals. Such corrosion shall be defined as any visible degradation of the equipment surfaces that can be

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attributed to flakey, pitted, blistered, or otherwise loosened finish of metal surface.

3.7.4.6 Fungus (see 4.7.5.6). The MK-1564( )/AIC assembly shall pass, if after 28 days, based on visual examination, it shows no more than only sparse microbial growth with restricted tuberal growth development in an area of 1 to 10 percent or less of the total area and no more than six unrelated minute colonies with mycelial development in areas only in other than critical circuit portions, such as terminal spacing, printed circuit boards, etc., with sparse growth due to random contamination or traces of unmixed material ingredients. The equipment shall fail, if it shows more than the growth specified above.

3.7.4.7 Blast (see 4.7.5.7). The MK-1564( )/AIC shall meet the requirements of 3.7.2.1.

3.7.4.8 Vibration (see 4.7.5.8). The MK-1564( )/AIC shall meet the requirements of 3.7.2.1.

3.7.4.9 Shock-drop (see 4.7.5.9). The MK-1564( )/AIC shall meet the requirements of 3.7.1 and 3.7.2.1. There shall be no evidence of breaking, cracking, or other structural damage.

3.7.4.10 Rain (see 4.7.5.10). The MK-1564( )/AIC shall meet the requirements of 3.7.2.1.

3.7.4.11 Overload. When tested as specified in 4.7.5.11 the frequency response of the earcup transducer assembly shall be within ± 3 dB of the requirements of paragraph 3.7.2.1.

3.7.4.12 Talk out. When tested as specified in 4.7.5.12 the microphone and earphones of the MK-1564( )/AIC shall be functioning properly and if so designed, there shall be only a minimal difference in the sound level output of the microphone when speaking directly into the front or back of the microphone.

3.7.5 Magnetic.

3.7.5.1 Effect of external magnetic field (see 4.7.6.1). The sensitivity of the MK-1564( )AIC to an external 400 Hertz (Hz) magnetic field shall not be more than 25 percent of that of a coil consisting of 123 turns of no. 34 enameled wire closely wound in a single-layer solenoid upon a nonconductive, nonmagnetic mandrel 0.500 inch in diameter, placed in the same magnetic field.

3.7.5.2 Stray magnetic field of the microphone (see 4.7.6.2). The stray magnetic field of the MK-1564( )/AIC shall cause no more than a 5-degree deflection of a magnetic compass at a distance of 12 inches.

3.7.5.3 Effects of electromagnetic interference (see 4.7.6.3). When tested as specified (see 3.1), the effects of electromagnetic interference on the MK-1564( )/AIC shall be in accordance with MIL-STD-461, in the following areas of concern:

a. Conducted emissions.
b. Radiated emissions.
c. Radiated susceptibility.

NOTE: Conducted susceptibility test does not apply to 2-wire devices because signal and power are on the same conductors.

3.8 Weight. The weight of the earcup transducer assembly shall not exceed 140 grams.

3.9 Color. The exterior surface of the microphone shall be colored semi-gloss black (27038), in accordance with FED-STD-595/27038. The circumaural earcups and the cables shall be black. Exterior finish shall be in accordance with MIL-DTL-14072.

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3.10 Identification of product. The identification marking shall include the following information only:

a. Microphone
MICROPHONE, LINEAR M-162/AIC
MANUFACTURER'S NAME OF FSCM
ORDER NO. *_____(Government order number)

b. Earphone element
EARPHONE H-347( )/AIC
MANUFACTURER'S NAME OR FSCM
ORDER NO. ____*_____(Government order number)

c. Circumaural earcup
EARCUP, CIRCUMAURAL MX-18199( )/AIC
MANUFACTURER'S NAME OR FSCM
ORDER NO. *_____(Government order number)

* Data to be inserted by the contractor per MIL-STD-1285. Marking shall be done in accordance with MIL-STD-1285. If MIL-STD-1285 does not cover a particular situation then MIL-STD-13231 shall apply.

3.11 Reliability (see 4.7.5.13). With a test duration of 0_1 = 3,000 hours and a duty cycle of 5 minutes "ON" to 1 minute "OFF", test conditions for temperature, vibration, and moisture should be as required by the Handbook (MIL-HDBK-781, see paragraph titled "Combined environments for turbopropeller aircraft and helicopter equipment") referenced in 6.4.

3.11.1 Reliability qualification test conditions. The reliability qualification test conditions should agree with the paragraph titled "Combined environments for turbopropeller aircraft and helicopter equipment", as required by the Handbook (MIL-HDBK-781) referenced in 6.4. A failure should be defined as the inability of the equipment to perform its required function. This includes catastrophic failures as well as any event which causes any departure from performance as required by this specification. Proper instrumentation should be provided to insure immediate recognition of failure. A failure of only one of the earphone elements is not considered a degradation or catastrophic failure.

3.12 Workmanship. The MK-1564( )/AIC should be manufactured and assembled in accordance with Guideline 9 of MIL-HDBK-454.

4. VERIFICATION

4.1 Responsibility for Inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all Inspection requirements as specified herein, Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such Inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with ISO-10012 and NCSL-Z540.3.

4.3 Classification of inspection. The inspections specified herein shall be classified as follows:

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a. First article inspection (see 4.5).
b. Conformance inspection (see 4.6).

4.4 Test conditions. Unless otherwise specified, the MK-1564( )/AIC shall be tested under the following conditions:

Temperature:	Room ambient, (+ 15 degrees C to + 35 degrees C).
Pressure:	Normal atmosphere.
Humidity:	Prevailing ambient up to 90 percent relative humidity.

4.4.1 Plotting of frequency response characteristic. The frequency response characteristics shall be plotted on a graphic level recorder (B K type 2305 or equivalent) using semi-logarithmic coordinate paper with the response in dB on the linear ordinate scale and the frequency in hertz (Hz) on the logarithmic abscissa scale. The length of a 100-to-1 frequency interval on the abscissa scale shall equal the length of 50 dB on the ordinate scale, (graphic level recorder paper B K Type QP1124 or equivalent). Or, at the option of the manufacturer, point-to-point measurements may be made every 50 Hz from 300 to 600 Hz; every 100 Hz from 600 to 1,500 Hz; every 250 Hz from 1,500 to 2,000 Hz; every 100 Hz from 2,000 to 2,500 Hz; every 250 Hz from 2,500 to 3,000 Hz; and every 500 Hz from 3,000 to 6,000 Hz, with the data recorded and a graph drawn and shall meet the requirements of 3.7.2.1 .

4.5 First article inspection. First article inspection shall be performed by the contractor after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. First article approval is valid only on the contract under which it is granted, unless extended by the Government to other contracts.

4.5.1 Sample size. Twenty (20) MK-1564( )/AIC shall be subjected to first article inspection. Also, 10 (ten) additional unterminated cables shall be provided for cable flex testing.

4.5.1.1 Test sample for the procuring activity. When first article inspections are conducted at a location other than the laboratory of the procuring activity, an additional quantity of 3 (three) untested samples shall be furnished to the procuring activity for such additional tests or examinations as may be desired.

4.5.2 Data to accompany test samples. Test sample quantities specified in 4.5.1 and 4.5.1.1 shall be accompanied by the following data.

a. A compilation of the extent of compliance with each requirement in section 3. This shall be done paragraph by paragraph with no omission.
b. A compilation of the contractors intentions with regard to any variation or deviation, which may appear in the compilation of 4.5.2.a.

4.5.3 Test program and report. When first article tests are to be conducted at a location other than the laboratory of the procuring activity, the contractor shall furnish the following:

a. Test program for approval.
b. Test report.

4.5.3.1 Test program. Prior to any formal first article testing, the contractor shall submit a Proposed First Article Test Program for approval consisting of the following items:

a. A list of all tests to be performed.
b. Complete procedures for each test to be performed, including block or schematic diagrams.
c. A list of test equipment to be used, identified by manufacturer and model number in case of standard test equipment or described by parameters and characteristics in the case of non-

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standard test equipment.

d. Copies of all data record forms to be used in recording and reporting test data.

4.5.3.2 Test report. For information about preparation of test reports in connection with completion of preproduction testing in accordance with an approved test program, see Handbook (MIL-HDBK-831) referenced in 6.5.

4.5.4 Inspection routine. The sample shall be subjected to the inspections specified in table IV, in the order shown. All sample units shall be subjected to the inspections of group I. The sample shall then be divided equally into two groups and subjected to the inspections for their particular group.

4.5.5 Failures. More than one failure in group I and any failure in group II or III shall be cause for refusal to grant first article approval.

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TABLE IV. First article inspection.

Inspection	Requirement Paragraph	Method Paragraph
Group I		
Visual and mechanical examination----------	3.4, 3.5, 3.6, 3.8 3.8, 3.10 and 3.12	4.7.1
Impedance----------	3.7.1.1	4.7.2.1
Insulation resistance----------	3.7.1.2	4.7.2.2
Dielectric withstanding voltage----------	3.7.1.3	4.7.2.3
Frequency response----------	3.7.2.1	4.7.3.1
Effect of external magnetic field----------	3.7.5.1	4.7.6.1
Stray magnetic field of the microphone----------	3.7.5.2	4.7.6.2
Talk out----------	3.7.4.12	4.7.5.12
Group II		
Psychoacoustic ----------	3.7.2.2	4.7.3.2
Harmonic distortion----------	3.7.2.3	4.7.3.2
Attenuation capability of earcup/transducer----		
Assembly----------	3.7.2.4	4.7.3.4
Real ear attenuation ----------	3.7.2.4.1	4.7.3.4.1
Flat plate coupler attenuation ----------	3.7.2.4.2	4.7.3.4.2
Linearity ----------	3.7.2.5	4.7.3.5
Cord, cordage and cable ----------		
Temperature range ----------	3.7.3.1	4.7.4.1
Flexing life -----(unterminated cable)---------	3.7.3.2	4.7.4.2
Cord anchorage ----------	3.7.3.3	4.7.4.3
Conductor strain relief and twist relief ------	3.7.3.4	4.7.4.4
Dielectric withstanding voltage of cord Assembly----------	3.7.3.5	4.7.4.5
Insulation resistance of cord assembly -----	3.7.3.6	4.7.4.6
Effects of electromagnetic interference----- (EMI)	3.7.5.3	4.7.6.3
Group III Environmental		
Subgroup A (2 samples)		
Temperature altitude----------	3.7.4.1	4.7.5.1
Humidity ----------	3.7.4.2	4.7.5.2
Blast ----------	3.7.4.7	4.7.5.7
Subgroup B (2 samples)		
Rain ----------	3.7.4.10	4.7.5.10
Salt fog ----------	3.7.4.5	4.7.5.5

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TABLE IV. First article inspection. Continued

Inspection	Requirement Paragraph	Method Paragraph
Group III Environmental		
	3.7.4.4	4.7.5.4
Subgroup C (2 samples)	3.7.4.8	4.7.5.8
Dust ----------	3.7.4.9	4.7.5.9
Vibration ----------	3.7.4.3	4.7.5.3
Shock drop ----------		
Immersion ----------		
	3.7.4.6	4.7.5.6
Subgroup D (2 samples)		
Fungus	3.11	4.7.5.13
Subgroup E (2 samples)		
Reliability		

4.6 Conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspections. Except as specified in 4.6.1.4.4, delivery of products which have passed groups A and B inspections shall not be delayed pending the results of group C inspection.

4.6.1.1 Inspection lot. An inspection lot shall consist of all units of the same type, produced under essentially the same conditions, and offered for inspection at one time.

4.6.1.2 Group A inspection. Group A inspection shall consist of the examination and test specified in table V, in the order shown.

4.6.1.2.1 Sampling plan. Accept on zero, for general inspection as specified in table VI, group A. Classification of major and minor defects for visual and mechanical examination is shown in table IX.

TABLE V. Group A inspection.

Inspection	Requirement Paragraph	Method Paragraph
Visual and mechanical examination	3.4, 3.5, 3.6, 3.8 3.9, 3.10 and 3.12	4.7.1
Impedance----------	3.7.1.1	4.7.2.1
Frequency response----------	3.7.2.1	4.7.3.1
Talk out----------	3.7.4.12	4.7.5.12

4.6.1.2.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as re-inspected lots.

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TABLE VI. Sampling plan.

Lot size	Sample size	
	Group A	Group B
2 to 8	100 percent	2
9 to 13	100 percent	3
14 to 25	13	3
26 to 50	13	5
51 to 90	13	6
91 to 150	13	7
151 to 280	20	10
281 to 500	29	11
501 to 1200	34	15
1201 to 3200	42	18
3201 to 10,000	50	22
10,001 to 35,000	60	29

4.6.1.3 Group B inspection. Group B inspection shall consist of the test specified in table VII and shall be made on sample units which have been subjected to and have passed group A inspection.

4.6.1.3.1 Sampling plan. Accept on zero, for general inspection as specified in table VI, group B. Classification of major and minor defects for visual and mechanical examination is shown in table IX.

TABLE VII. Group B inspection.

Inspection	Requirement Paragraph	Method Paragraph
Linearity----------	3.7.2.5	4.7.3.5
Temperature range ----------	3.7.3.1	4.7.4.1
Flexing life – (unterminated cable)----------	3.7.3.2	4.7.4.2
Conductor strain pellet and twist relief ----------	3.7.3.4	4.7.4.4

4.6.1.3.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as re-inspected lots.

4.6.1.3.3 Disposition of sample units. Sample units which have passed the group B inspection may be delivered on the contract or purchase order, if the lot is accepted and the sample units are still within specified electrical tolerances, and provided the cord assemblies of the tested units are replaced prior to delivery.

4.6.1.4 Group C inspection. Group C inspection shall consist of the tests specified in table VIII, in the order shown. Group C inspection shall be made on sample units selected from inspection lots which have passed groups A and B inspections.

4.6.1.4.1 Sampling plan.

4.6.1.4.1.1 Ten (10) samples of the complete kit shall be selected at random, 5 (five) samples for each subgroup 1 and subgroup 2 in table VIII. The samples shall be selected at the start of the contract from the first Conformance Inspection lot. These samples shall constitute the group C requirement for the first 1,000 units produced.

4.6.1.4.1.2 Thereafter, 3 (three) samples of the complete kit shall be for subgroup 1 in table VIII. These

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samples shall be selected once each month, or every 1,000 units, whichever occurs first.

TABLE VIII. Group C inspection.

Inspection	Requirement Paragraph	Method Paragraph
Subgroup 1 (5 sample units)		
Psychoacoustic	3.7.2.2	4.7.3.2
Harmonic distortion	3.7.2.3	4.7.3.3
Humidity	3.7.4.2	4.7.5.2
Dust	3.7.4.4	4.7.5.4
Vibration	3.7.4.8	4.7.5.8
Attenuation		
Subgroup 2 (5 sample units)		
Sub-Subgroup A (2 samples)		
Temperature-altitude	3.7.4.1	4.7.5.1
Blast	3.7.4.7	4.7.5.7
Sub-subgroup B (2 samples)		
Rain	3.7.4.10	4.7.5.10
Salt fog	3.7.4.5	4.7.5.5
Overload	3.7.4.11	4.7.5.11
Sub-Subgroup C (1 sample)		
Shock drop	3.7.4.9	4.7.5.9
Immersion	3.7.4.3	4.7.5.3
Fungus	3.7.4.6	4.7.5.6

4.6.1.4.2 Failures. If one or more sample units fail to pass group inspection, the sample shall be considered to have failed.

4.6.1.4.3 Disposition of sample units. Sample units which have been subjected to group C inspection shall not be delivered on the contract or purchase order.

4.6.1.4.4 Noncompliance. If a sample fails to pass group C inspection, the supplier shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspections, or the inspection which the original sample failed, at the option of the Government). Groups A and B inspections may be reinstituted; however, final acceptance shall be withheld until the group C reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and corrective action taken shall be furnished to the responsible inspection activity and the procuring activity.

4.6.1.4.5 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and the method of correcting them.

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4.7 Methods of examination and test.

4.7.1 Visual and mechanical examination. The MK-1564( )/AIC shall be examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.4, 3.5, 3.6, 3.8, 3.9, 3.10, and 3.12). Unless otherwise specified (see 6.2), the defects shall be classified as specified in table IX. Particular attention shall be paid to the following:

a. Soldering operations.
b. Shield.
c. Diaphragm.
d. Uniformity and strength of cementing.
e. Connections to FET amplifier of microphone.
f. Security and cleanliness of microphone and earphone terminal connections to cable terminals.
g. Construction of cord assembly kit.
h. Vent in microphone case and earphone cases for pressure equalization.
i. Cleanliness of vent holes throughout the microphone and earphone structure.
j. Workmanship regarding the fit and sealing of the microphone and earphone cases and fit and sealing of the moisture barrier to the microphone and earphones.
k. Set screws holding the cables to the microphone and earphones. (They shall be slotted, headless screws.)

4.7.2 Electrical performance of the earcup transducer assembly.

4.7.2.1 Impedance (see 3.7.1.1). The impedance of the assembly shall be measured with 0.13 volts, 1 kHz signal applied to the earphone terminals. The impedance shall be determined either by measuring the voltage across and the current through the earphone or with an impedance bridge.

4.7.2.2 Insulation resistance (see 3.7.1.2). The assembly shall be tested in accordance with method 302, test condition B, of MIL-STD-202. There shall be no evidence of arcing or insulation breakdown during the test.

4.7.2.3 Dielectric withstanding voltage (see 3.7.1.3). The assembly shall be tested in accordance with method 301 of MIL-STD-202. The following details shall apply:

a. Test voltage: 100 volts.
b. Nature of potential: 60 Hz ac.
c. Points of application Between one terminal and the external surface of the earphone case.
d. Measurements after test; Impedance at 1,000 Hz.

4.7.3 Acoustical performance of the earcup transducer assembly.

4.7.3.1 Frequency response (see 3.7.2.1). The frequency response of the earphone in an earcup shall be made starting at 300 Hz and extending through 6,000 Hz. The earcup transducer assembly shall be placed on a flat plate coupler of an artificial ear, B&K Type 4153 or demonstrated equivalent, with a 1 Kg force (to simulate closure to the head). A pure tone sine wave from an audio oscillator (B&K 1024 or equal) at 105 dB SPL (Re: 20 micropascal's (µPa) measured by the artificial ear will be used as the input to the device under test. The frequency response of the earcup transducer assembly shall be in accordance with paragraph 3.7.2.1. NOTE: In addition to the flat plate coupler described above, as a final check on earcup transducer design real head frequency response measurements shall be made utilizing a miniature electret microphone placed on the conchae of the human head.

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4.7.3.2 Psychoacoustlc (see 3.7.2.2). The MK-1564( )/AIC shall be subjected to a series of psychoacoustic tests, using the Harvard phonetically balanced words in accordance with ANSI S3.2. The noise environment for the talkers shall be pink noise at 100 dB SPL presented in a diffuse field environment, as described in ANSI Standard S12.6. The listeners shall be in the same environment and wearing SPH-4 Helmets, with the MK-1564( )/AIC Installed.

4.7.3.3 Harmonic distortion (see 3.7.2.3). The earcup transducer assembly shall be mounted on the flat plate coupler, and connected to the test circuit described in 4.7.3.1. The output of the oscillator shall be adjusted to 0.26 volts at 300 Hz across the series combination of the 19 ohm resistor and the earphone. The rms harmonic distortion shall be measured with a Hewlett-Packard Model 334-A Distortion Analyzer or equal, connected across the output of the microphone referenced in 4.7.3.1. The distortion measurement shall be repeated at 400 Hz and at sufficient points between 400 Hz and 5,000 Hz to determine the frequency where maximum distortion exists. The total harmonic distortion shall comply with 3.7.2.3.

4.7.3.4 Attenuation capability of earcup transducer assembly.

4.7.3.4.1 Real head - Attenuation (see 3.7.2.4.1). The earcup transducer assembly shall be mounted on a human head, with a miniature electret microphone placed on the concha of that ear. A B&K ½ inch microphone shall be placed next to the assembly, to measure the noise outside the earcup. The miniature electret microphone, including its amplifier, shall have a frequency response characteristic equivalent to the ½ inch B&K microphone and be of equal sensitivity. It will measure the noise inside the earcup. The difference of these values shall be measured at every 1/3 octave frequency between 100 Hz and 10 kHz in a free field 105 dB SPL pink noise environment. A minimum sample of 10 individuals of varying head sizes shall be tested and a composite average be made. This average will be the real head attenuation of the earcup transducer assembly. Measurements will be recorded using a sweeping 1/3 octave band analyzer or in tabular form. Where ANSI S12.6 conflicts with this specification, this specification shall govern.

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TABLE IX. Classification of defects for visual and mechanical examination.

Defect type	Classification	
	Major	Minor
Dimensions	Dimensions not as specified	-----
Materials and finish	Materials not as specified. Wrong or incomplete finish. Large amounts of flaking, peeling, or chipping of finish.	Scratches, cuts, abrasions, etc., causing exposure of base metal, or relatively small amounts of flaking peeling, or chipping.
Parts	Missing parts. Inoperative, improperly assembled, or defective parts, which could cause the unit to fail in service. Wrong parts.	Defective parts which would reduce efficiency of use, but not cause failure in service. Cracks or chipped surfaces having no effect on the functioning, assembly, maintenance, or life of the unit.
Marking	Marking missing, illegible, or incorrect.	Markings dirty or smudged, but legible.
Foreign objects	Any metallic foreign object, not firmly attached[1/], which could cause a short circuit, or acoustical malfunctioning of the unit. Any nonmetallic foreign object, such as insulation, dirt, or phenolic chips which could cause acoustical malfunctioning of the unit.	Any metallic or nonmetallic foreign object, which affects appearance, but which could not cause acoustical malfunctioning of the unit.
Soldering	Improper wrap – less than ½ turn Unsoldered joint – solder not applied where intended. Insufficient solder – Minimum dimension of solder bridge less than twice the diameter of the wire or less than 3/32 inch, whichever is greater. Entire area of contact between wire and terminal not jointed by solder bridge. Cold solder joint – chalky appearance,. Lacks metallic luster, presents rough “pile-up” appearance, movement of wire or solder upon pick application. Rosin joint – Presence of excess rosin; relative movement of wire or solder upon pick application. Insulation in terminal hole – solder over insulation, nor appearance of visible wire contour.	Improper wrap – ½ turn or more, but less than one turn. Excess solder – build-up of solder on joint greater than necessary for good soldering, usually resulting in obliteration of wire contour. Cold solder joint – chalky appearance lacks metallic luster, presents rough “pile-up” appearance, no relative action between wire and solder upon pick application.
Wiring	Wiring not in accordance with schematic diagram. Broken strands – More than 20 percent; except in a 7-strand conductor, more than 2 broken strands. Insulation burned, abrased, pinched, or deteriorated between two or more conductors, resulting in a potential short circuit. Taut wire – Wire exhibits no slack and subsequent breakage may occur due to stress on terminal or part. Insulation frayed to the extent that a potential short circuit exists.	Broken strands – 20 percent or less. In a 7-strand conductor, 2 broken strands or less. Insulation burned, abrased, pinched, or deteriorated, with exposure of bare wire, but short circuit not possible Taut wire – Slight stress on conductor, but no possibility of subsequent breakage.

1/ Foreign objects that cannot be dislodged by the moderate application of pressure with a prick or spudger shall be considered to be firmly attached.

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4.7.3.4.2 Flat plate coupler attenuation (see 3.7.2.4.2). A B&K Type 4153 artificial ear with flat plate (or equivalent shall be placed in a free field of 105 dB SPL pink noise measured by the artificial ear's microphone. The 1/3 octave values measured from 200 Hz to 5 kHz shall be recorded. Next the earcup transducer assembly is placed on the flat plate and a 1 kg force placed on the outside of the assembly to simulate closure to the head. A new set of 1/3 octave values shall be measured and recorded. The difference between these values shall be the flat plate coupler attenuation of the assembly. Measurements will be recorded using a 1/3 octave band analyzer or in tabular form.

4.7.3.5 Linearity (see 3.7.2.5). The input voltage to the assembly shall be adjusted to produce an output measured across a 19 ohm resistive load equal to 85 dB, which is relative to 20 micropascal's (µPa) at any one frequency within each of the following ranges: 400-800 Hz, 800-2.500 Hz, and 2,500-4,000 Hz. The input voltage measured, which produces 85 dB re 20 µPa, at the selected frequency in each of the listed ranges, will be increased by 10 dB and 20 dB and the acoustic output of the assembly shall be checked for a respective increase in output level of 10 dB and 20 dB $\pm$ 1 dB at each of the selected frequencies.

4.7.4 Cord, cordage, and cable.

4.7.4.1 Temperature range (see 3.7.3.1). The unterminated cable shall be placed within a test chamber, wherein a temperature of + 85 degrees C (+ 185 degrees F) is maintained for a period of 4 hours. The cable shall be wrapped around a mandrel while at this temperature without damage to the cable jacket or conductor insulation. The diameter of the mandrel shall be the same as the nominal cable diameter. The cable shall be attached to the mandrel and suspended vertically with the lower end weighted sufficiently to keep the specimen taut and to permit wrapping without handling. The cable shall be inspected for damage or deformation and then subjected to a test chamber wherein the temperature of -55 degrees C (-67 degrees F} is maintained. After 20 hours of storage at -55 degrees C, the cable shall be wrapped about the mandrel while at this temperature, in the same manner as it was at the positive extreme temperature. The cable shall be examined for cracks or other damage to the jacket or conductor insulation.

4.7.4.2 Flexing life (see 3.7.3.2). Five samples of unterminated cable, each approximately 1-1/2 feet long, shall be subjected to the following test. Other flexing test methods which will subject the cable to flexing cycles equivalent to flexing cycles specified herein may be used upon approval by the procuring activity. The sample lengths shall be clamped and suspended through holes in a rectangular (1/2 by 5/8 inch) metal bar in the manner described below:

a. The clamping point shall be immediately above the metal bar, the cable clamp shall rest on top of the metal bar.
b. The cable shall be clamped in such a manner that it will not turn in the hole during the flexing test.
c. The length of the through holes in the metal bar shall be 5/8 inch. The diameter of the through holes shall be 0.005 to 0.010 inch greater than the diameter of the cable undergoing test.
d. The through holes in the metal bar shall incorporate a 1/8 inch radius at the end from which the cable emerges (bending point).
e. A knot shall be tied near the free end of the cable and a 1/2 pound weight shall be hung from the knot.
f. The four center conductors plus shield of each sample length of cable, a small low-current drain, 6-volt/150 mA lamp, and a suitable power source shall be connected in series to indicate electrical continuity. The metal bar, supported horizontally, shall then be rotated about its longitudinal axis back and forth through an angle of 120 degrees (60 degrees each side of vertical) at a rate of approximately 85 cycles per minute. The arithmetical average of the number of flexes of each of the 5 sample lengths of cable, before electrical discontinuity occurs, shall be taken as the flexing life of the cable.

4.7.4.3 Cord anchorage (see 3.7.3.3). A pull of 8 pounds shall be applied to the cable in the direction

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of the axis of the connector plug for a minimum of 30 minutes. The cord assembly shall be examined to determine whether any slippage has occurred at the clamp and if any strain has been exerted on the conductors.

4.7.4.4 Conductor strain relief and twist relief (see 3.7.3.4). With the pins of one connector plug anchored rigidly in an earphone and in a vertical position, a 5-pound weight shall be suspended from the pins of the other connector plug of the cord assembly for a period of at least 30 minutes. During the last 5 minutes of the pull test, the lower plug connector with the 5-pound weight suspended shall be twisted 3 turns clockwise and 3 turns counter-clockwise from the normal resting position. The twisting cycle shall be repeated 3 times. The impedance of the microphone and earphones with cord assembly in place shall be measured at 1,000 Hz and a mechanical inspection shall be made before and immediately upon completion of the test. A dielectric withstanding voltage test shall be made on the cord assembly in accordance with the procedure specified in 4.7.4.5, after the impedance measurement and mechanical inspection have been made.

4.7.4.5 Dielectric withstanding voltage of cord assembly (see 3.7.3.5). The dielectric of the finished cord assembly shall be determined in accordance with MIL-DTL-55668.

4.7.4.6 Insulation resistance of cord assembly (see 3.7.3.6). The insulation resistance of the finished cord assembly shall be determined in accordance with MIL-DTL-55668.

4.7.5 Environmental.

4.7.5.1 Temperature altitude (see 3.7.4.1). The unit under test shall be subjected to MIL-STD-810, Revision B, Method 504, Procedure I, for Class 1B equipment in accordance with table X. It is shown here (in part) as it appeared in MIL-STD-810 (Revision B), with the exceptions that the temperature specified for step 1 and 5 shall be changed to -57 degrees and +71 degrees C, respectively. Steps 2, 4, 7, 8, 10, 12, and 13 shall be omitted. After each step, operation of the equipment shall meet the requirements of 3.7.2.1. In addition, the procedure shall include a one-hour exposure (non-operating) to a pressure of 5.6 inches of mercury (2.75 PSI), height simulating air transport at 40,000 feet.

Method 504 Temperature-Altitude was reconfigured within MIL-STD-810C.
THE CONTENTS OF THE ORIGINAL (CLASS 1B) TEST METHOD ARE INCLUDED AND TAILORED BELOW, TO ALLOW FOR A SIMPLIFIED METHOD OF TESTING.

4.7.5.1.1 Purpose. The temperature-altitude test is conducted to determine the ability of equipment to operate satisfactorily under simultaneously applied varying conditions of low pressure and high/low temperature. The equipment category (Class 1B), as used in this method, is determined by the required altitude (15,000 feet) operating range and the required sea level continuous operating temperature (-62 degrees C to +85 degrees C).

4.7.5.1.1.1 General effects. Deleterious effects to be anticipated include cracking or rupture of materials due to contraction or expansion, short circuiting of electrical wiring, and other damaging effects which might be expected from exposure to any of the above environments.

4.7.5.1.2 Apparatus. Temperature-altitude chamber and auxiliary thermal sensors with associated recording devices [see note (a) at the end of 4.7.5.1.3.1]

4.7.5.1.3 Procedures.

4.7.5.1.3.1 Procedure I. A record of specific pretest data shall be made to determine that the test item performs within prime item specification requirements. The pretest shall also include the functional parameters to be monitored during and after the test, if not specified in the equipment specification. This shall include acceptable functional limits (with permissible degradation) when operation of the test item is

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required. In general, the testing schedule outlined in table X shall be followed. However, each step in table X represents a condition which the test item may encounter in service; therefore, each step may be applied independently of the others. For operating conditions other than those specified in 4.7.5.1, the alternate temperature-altitude conditions in figure 3 shall be used. When changing chamber conditions from those required for one step to those required for any other step, in the sequence given in table X or in any sequence, the rates of temperature and pressure changes may be the maximum attainable by the chamber. But, these rates shall not exceed 1 degree C (1.8 degrees F) per second for airborne equipment or 10 degrees C (18 degrees F) per minute for test equipment and 0.5 inch of mercury (0.24 PSI) per second. Pressures for altitude tables are contained in the latest addition of the U. S. Standard Atmosphere supplement.

TABLE X. Test conditions for temperature-altitude tests

Category	Step →	1a	1b	3	5	6	9	11	14
1B	Temp (°C)→	+25	-57	-40	+71	+85	+52	+49	Site 1/
	ALT (Feet)→	40,000	Site 1/	15,000	Site 1/	Site 1/	10,000	15,000	Site 1/
	Time (HR)→	1	2	---	16	4	4	4	---

1/ Site – Temperature and Altitude of test facility.

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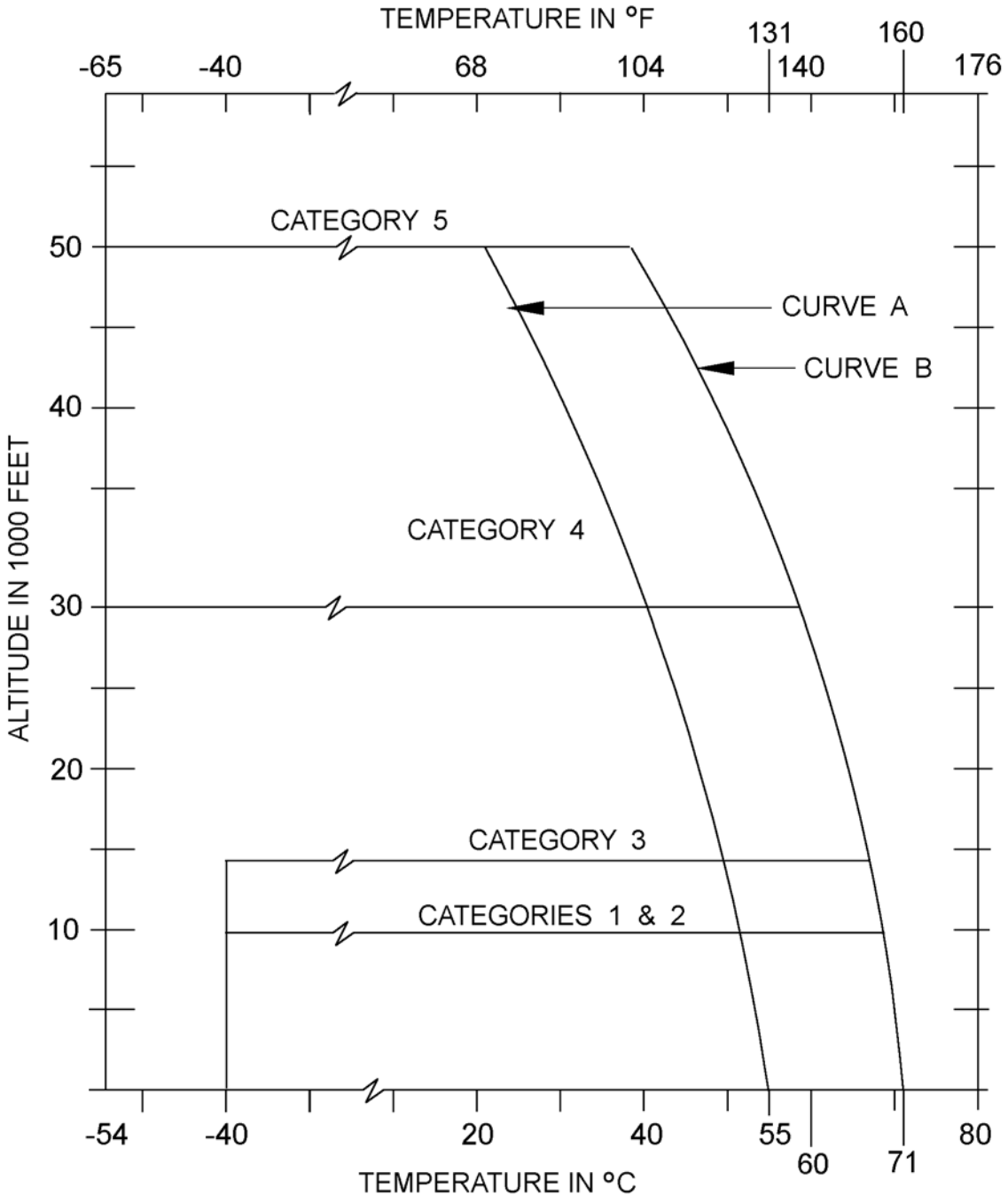


FIGURE 3 Temperature versus altitude operational requirements for Categories 1 through 5 equipment.

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Step 1a -

With the test item de-energized, adjust the chamber temperature to that specified in step 1a of table 3. After the test item temperature has stabilized, the chamber pressure shall be adjusted to simulate the altitude specified in step 1a of table 3. Maintain these test conditions for at least the length of time specified in step 1a of table 3. At the conclusion of this time period, the test item shall, to the extent practical, be visually inspected for the presence of deterioration which would impair future operation.

Step 1b -

With the test item de-energized, adjust the chamber test conditions to those specified in step 1b of table 3. After the test item temperature has stabilized, maintain these test conditions for at least the length of time specified in step 1b of table 3. Where it is possible, without changing the test conditions, the test item shall be visually inspected for the presence of deterioration which would impair future operation.

Step 3 –

With the test item de-energized, adjust the chamber temperature to that specified in step 3 of table X. The test item temperature shall be stabilized. The test item shall be energized at the highest specified input voltage (see note (d)) and the chamber pressure adjusted to simulate the altitude specified in step 3 of table X. Upon reaching the specified chamber pressure and while maintaining the chamber temperature, the test item shall be checked for satisfactory operation, the results recorded, and evaluated using the failure criteria of 3.7.2.1 .

Step 5 –

Adjust the chamber test pressure conditions to standard sea-level conditions and the temperature to that specified in step 5 of table X. After the test item temperature has stabilized; an operational and performance check of the test item shall be made and the results compared with the data obtained during pretest and evaluated using the failure criteria of 3.7.2.1.

Step 6 –

With the test item de-energized, adjust the chamber test conditions to those specified in step 6 of table X. The test item temperature shall be stabilized and then maintained for at least the length of time specified in step 6 of table X. At the conclusion of this time period, where practicable, the test item shall be visually inspected for any deterioration.

Step 9 –

With the test item non-operating, adjust the chamber temperature to that specified for step 9 of table X. The test item temperature shall then be stabilized. The test item shall then be turned on at the highest specified input voltage and the altitude shall be adjusted to that specified. Following altitude and test item temperature stabilization, the test item shall be operated at the highest specified input voltage for 4 hours. Thermocouple readings of the test item shall be recorded every 30 minutes. At the end of the specified operating period, continue to operate the test item until the equipment has been checked for satisfactory operation; results recorded and evaluated using the failure criteria of 3.7.2.1.

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Step 11 –

With the test item non-operating, adjust the chamber temperature to that specified for step 11 in table X. Following chamber temperature adjustment, the test item shall be turned on and the altitude adjusted to that specified. After the chamber conditions have stabilized, permit the test item to operate at the highest specified input voltage for 4 hours. Thermocouple readings of the test item shall be recorded every 30 minutes. At the end of the specified operating period, continue to operate the test item at the specified conditions until an operational and performance check is made, the results recorded, and evaluated using the failure criteria of 3.7.2.1.

Step 14 –

With the test item operating, adjust the chamber conditions to standard ambient conditions. When the chamber conditions have stabilized, an operational and performance check shall be made on the test item and evaluated using the failure criteria of 3.7.2.1.

Notes:

a. The following guidelines are provided for consideration when determining location of thermal sensors used to monitor the test items:

1. One or more sensors in the ambient air within each major unit.

2. Contact temperature on the largest mass in each major unit.

3. Contact temperature on the part(s) where the highest surface temperature is expected.

4. Contact temperature on the part(s), whose temperature is likely to limit equipment performance.

b. When the chamber door is opened it is intended that frost will form; however, should the relative humidity of the air be such that frost will not form, an artificial means shall be used to provide the relative humidity necessary to have frost form.

c. Following those steps where a change in temperature at low pressure is required, the pressure may be increased to ambient before changing the temperature and then returned to the required pressure following temperature stabilization.

4.7.5.2 Humidity (see 3.7.4.2). The equipment shall be capable of meeting the test of Method 507, Procedure II, of MIL-STD-810. The equipment shall meet the requirements of 3.7.2.1 and 3.7.1.1.

4.7.5.3 Immersion (see 3.7.4.3). The equipment shall be capable of meeting the test of Method 512, Procedure I, of MIL-STD-810. Within one (1) hour following the immersion period, the equipment shall meet the requirements of 3.7.2.1.

4.7.5.4 Dust (see 3.7.4.4). The equipment shall be capable of meeting the test of Method 510, Procedure I, of MIL-STD-810. The equipment shall meet the requirements of 3.7.2.1.

4.7.5.5 Salt fog (see 3.7.4.5). The equipment shall be capable of meeting the test of Method 509, Procedure I, of MIL-STD-810. Operation of the equipment shall not be required. Following the rest, the equipment shall meet the requirements of 3.7.1.1 and 3.7.2.1.

4.7.5.6 Fungus (see 3.7.4.6). The equipment mode) in the assembled and as ready for delivery condition, shall be capable of meeting 28 day test of Method 508, MIL-STD-810. The equipment test

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model shall not be specially cleaned for the fungus test, except for the cleaning It receives during or after production. If It Is necessary for the manufacturer after assembly to remove accumulated production and handling contaminants by cleaning prior to packaging and delivery, the equipment shall not he tested for three days after cleaning to allow for complete evaporation of the cleaning compound. All enclosed or gasketed assemblies shall be opened and the interior sprayed with the specific mixed spore suspension. After 14 and 28 days of test, the control items per Method 508, MIL-STD-810 shall show profuse growth over at least 50 percent of the area of the control items. The test shall be repeated if the control Items fail to show profuse growth after 14 and 28 days of test.

4.7.5.7 Blast (see 3.7.4.7). Each MK-1564( )/AIC being tested shall be mounted on the carriage of the U.S. Navy Simulated Gun Blast Equipment in accordance with Bureau of Ships Drawing SK-N-864 with the front edge of the earphone element in the test plane, and with its axis coincident with that of the explosion chamber. The earphone element shall be subjected to 30 rounds of blast at a nominal peak pressure of 9.5 pounds per square inch.

4.7.5.8 Vibration (see 3.7.4.8). The test item shall be attached to the vibration exciter table in a horizontal position, with the applied vibration in a direction perpendicular to the plane of the microphone. Subject the equipment to sinusoidal vibration over a frequency range of 5 to 20 Hz at a constant displacement (double amplitude) of 0.10 inches, and over a frequency range of 20 to 500 Hz at a constant acceleration of 2 G (see figure 4, curve B, from MIL-STD-810C, Method 514, dated 10 March 1975). The frequency of the applied vibration shall be swept logarithmically over the specified frequency range. One vibration cycle (5-500-5) shall consume 15 minutes. Total time of test shall be 2 hours. Repeat the above vibration in the fore/aft and side to side axis. Upon completion of the test, the equipment shall be undamaged and shall meet the requirements of 3.7.2.1.

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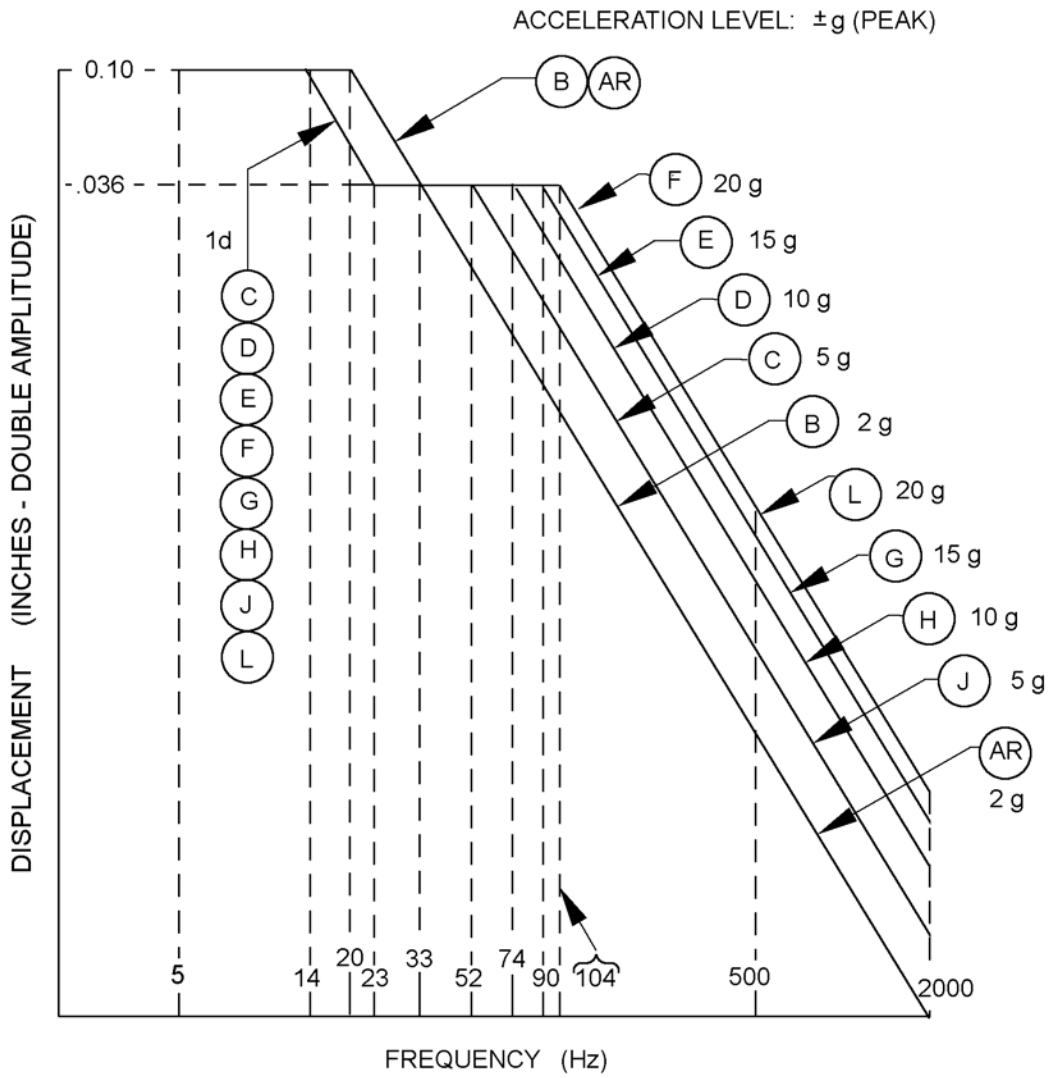


FIGURE 4. Vibration Test Curves for Equipment installed in airplanes, excluding helicopters, equipment category b.1.

4.7.5.9 Shock, drop (see 3.7.4.9). The MK-1564( )/AIC shall be dropped at random from a height of 6 feet, 12 times onto a concrete surface. At completion of the test, the microphone and earphones shall meet the requirements of 3.7.1 and 3.7.2.1. There shall be no evidence of breaking, cracking or other structural damage.

4.7.5.10 Rain (see 3.7.4.10). The equipment shall be subjected to the test of Method 506, Procedure I, of MIL-STD-810. The equipment shall be operated during the last 10 minutes of the 30 minute period. The tester shall determine compliance with paragraph 3.7.4.10.

4.7.5.11 Overload (see 3.7.4.11). The earcup transducer assembly shall be subjected to 8 hours of operation with 2.3 volts input at 1,000 Hz. At the end of 8 hours the assembly shall meet the requirements of 3.7.2.1.

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4.7.5.12 Talk out (see 3.7.4.12). The MK-1564( )/AIC shall be connected to a suitable amplifier with side tone included. The microphone shall also be connected to a 10 Vdc power supply or battery to power its internal amplifier. The tester shall wear the MK-1564( )/AIC and with lips touching the microphone, shall speak directly into the microphone (front and back sound ports if so designed) and listen to the earphones. The amplifier that the MK-1564( )/AIC is connected to shall not utilize automatic gain control (AGC) or other circuits which will compensate for low microphone output. The tester shall determine compliance with paragraph 3.7.4.12.

4.7.5.13 Reliability (see 3.11).

4.7.5.13.1 Reliability test procedures. For information about Reliability of the MK-1564( )/AIC see the test plan XVII D, in the Handbook (MIL-HDBK-781) referenced in 6.4 using a minimum sample of two (2) equipment's.

4.7.5.13.2 Parameters to be measured. As a minimum, Frequency Response (see 3.7.2.1) and Harmonic Distortion (see 3.7.2.3) should be measured and recorded daily in agreement with the Handbook referenced in 6.4, with the exception that a complete set of measurements should be taken daily during both the high and the low temperature stabilization. Vibration should not be avoided during measurement. Parameters to be measured should be those included in the contractor's approved reliability test procedure.

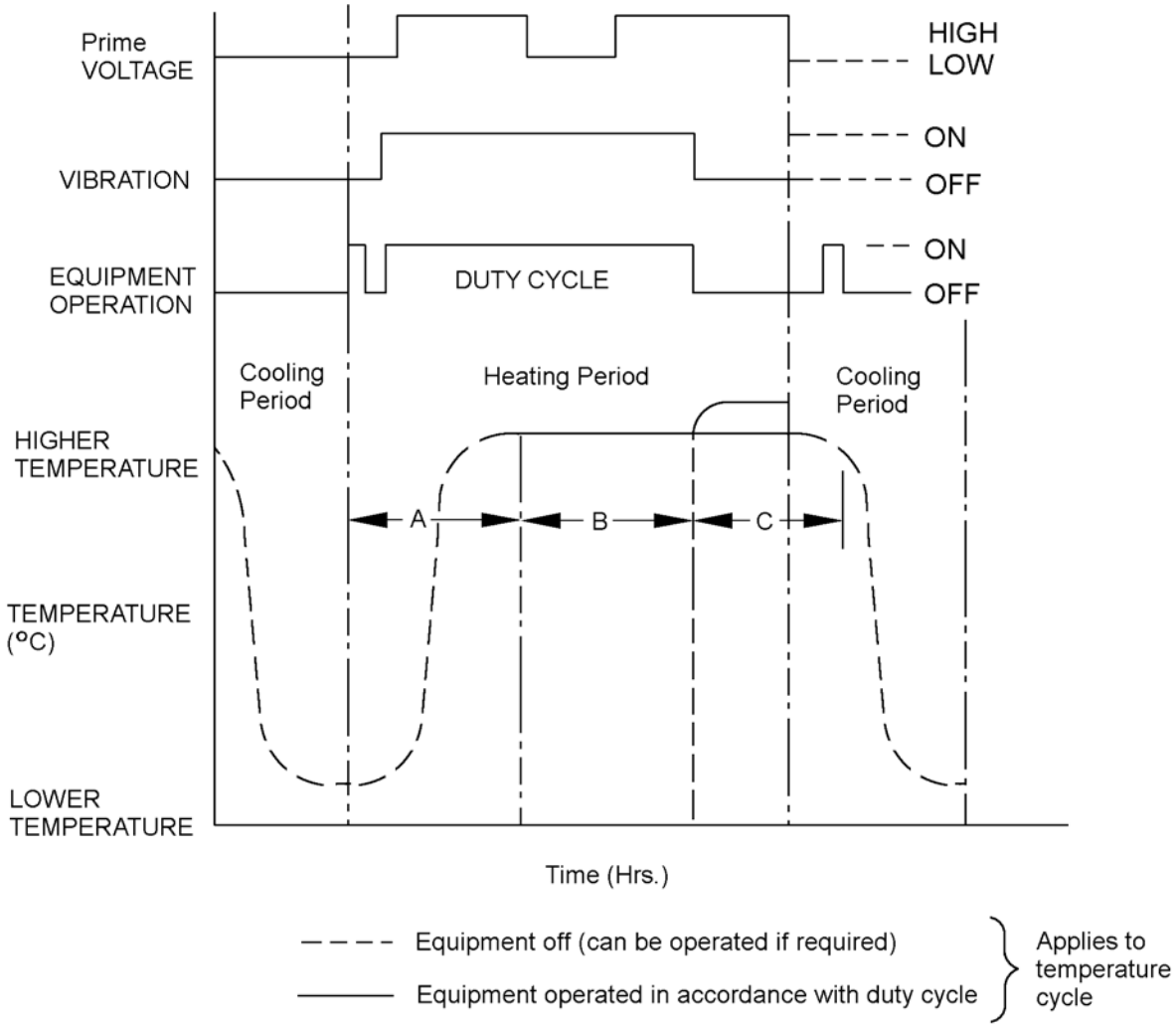
Parameter	Requirement Paragraph
Frequency Response	3.7.2.1
Harmonic Distortion	3.7.2.3

The equipment performance parameters to be measured and the frequency of measurement should be specified in the test procedures. If the value of any required performance parameter is not within specified limits, a failure should be recorded. If the exact time of failure cannot be determined, the failure should be presumed to have occurred at the last recorded observation or successful measurement of that same parameter. Observations and measurements should be made and recorded during different portions of the test cycle (see A and B of figure 5) with at least one set of measurements recorded when the equipment is first energized after any specified equipment start-up period.

4.7.5.13.3 Rework and retest provisions. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units and resubmit for re-inspection. If one or more defects are found in the second sample, the lot shall be rejected and shall not be supplied to this specification. Such lots shall be separate from new lots and shall be clearly identified as re-inspected lots.

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A. Time for chamber to reach stabilization at higher temperature

B. Time of equipment opertion at higher temperature

C. Optional Hot Soak and hot start-up checkout

FIGURE 5. Sample environmental test cycle.

4.7.6 Magnetic.

4.7.6.1 Effect of external magnetic field (see 3.7.5.1). The sensitivity of the MK-1564( )/AIC to an external ac magnetic field shall be measured by comparing the induced open-circuit voltage across the terminals of the microphone and earphones when placed in a 400 Hz magnetic field with the inducted open-circuit voltage across the terminals of a calibrating coil of known sensitivity placed in the same

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magnetic field.

4.7.6.1.1 Field coil. The magnetic field for this test may be produced by a coil of about 700 turns of No. 18 enameled wire 6 inches by 6 inches in cross section or 6 inches in diameter and winding length of 10 inches.

4.7.6.1.2 Calibrating coil. The calibrating coil shall consist of a single-layer solenoid wound with 123 turns of No. 34 enameled wire on a non-conductive, non-magnetic mandrel 0.5 inch in diameter.

4.7.6.1.3 Measurement procedure. Excite the field coil with about 250 milliamperes of 400 Hz alternating current. Connect the microphone and earphones to an electronic voltmeter. Closely twist the leads from the microphone and earphones to the voltmeter. Orient the microphone and earphones in the center of the field coil for maximum output voltage and record this value, then substitute the calibrating coil for the microphone and earphones. Orient the calibrating coil for maximum output voltage in that portion of the field coil previously occupied by the microphone and earphones and record the output voltage.

4.7.6.2 Stray magnetic field of MK-1564( )/AIC (see 3.7.5.2). The stray magnetic field of the MK-1564( )/AIC shall be determined as follows, using a Keuffel and Esser Type No. 5600 compass, or its approved equivalent. The microphone and earphone transducers shall be placed with their geometric center 12 inches from the pivot point of the compass needle and on the perpendicular bisector of the needle in the plane of rotation of the needle. The microphone and earphones shall be oriented in all directions and the maximum deflection of the compass observed. This test shall be made in a location substantially free from stray magnetic disturbances.

4.7.6.3 Effects of electromagnetic interference (EMI) (see 3.7.5.3). The MK-1564( )/AIC shall be subjected to the following test procedures of MIL-STD-461:

a. Conducted emissions - Method – CE101
b. Radiated emissions - Method – RE102
c. Radiated susceptibility - Method – RS101.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The MK-1564( )/AIC contains a linear earcup transducer assembly with improved far field noise attenuation properties over the MK-896A/AIC, it also contains the microphone, Linear M-162/AIC which is a noise-canceling dc powered, high gain microphone. The MK-1564( )/AIC is intended for use in the improved SPH.4 aviator’s helmet which is part of the airborne communications system which will provide communications of high intelligibility under the extreme noise conditions encountered in military aircraft.

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6.1.1 Packaging requirements. The preservation, packing and marking specified herein are intended for direct shipments to the Government. However, at the option of the contractor or when so specified, the packaging provisions herein are also applicable for the preparation of headset-microphone kits for shipment from the parts contractor to the original equipment manufacturer.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of the specification.
b. When first article inspection is required.
c. Packaging requirements (see 5.1).

6.3 Engineering data. Headset Microphone Kit MK-1564( )/AIC replaces the MK-896/AIC when used with the new aircraft intercom, C-10414/ARC. (It is not electrically interchangeable with the MK-896A/AIC.)

6.4 Reliability Test Methods (see 3.11). See MIL-HDBK-781, for information about “Reliability Test Methods, Plans, and Environments for Engineering Development, Qualification, and Production”.

6.5 Preparation of Test Reports (see 4.7.5.13). For information about “Preparation of Test Reports”, see MIL-HDBK-831.

6.6 Tin whisker growth (see 3.5.4). The use of (Tin/Lead) alloys with tin content greater than 97 percent, by mass, may exhibit tin whisker growth problems after manufacture. Tin whiskers may occur anytime from a day to years after manufacture and can develop under typical operating conditions, on products that use such materials. Conformal coatings applied over top of a whisker-prone surface will not prevent the formation of tin whiskers. Alloys of 3 percent lead, by mass, have shown to inhibit the growth of tin whiskers. For additional information on this matter, refer to ASTM B 545 (Standard Specification for Electrodeposited Coatings of Tin). Copies of this document are available from http://www.astm.org or ASTM International, P.O. Box C700, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

6.7 Environmentally preferable material. Environmentally preferable materials should be used to the maximum extent possible to meet the requirements of this specification. As of the dating of this document, the U.S. Environmental Protection Agency (EPA) is focusing efforts on reducing 31 priority chemicals. The list of chemicals and additional information is available on their website http://www.epa.gov/osw/hazard/wastemin/priority.htm. Included in the EPA list of 31 priority chemicals are cadmium, lead, and mercury. Use of these materials should be minimized or eliminated unless needed to meet the requirements specified herein (see 3.2).

6.8 Subject term (key word) listing.

Acoustical
Attenuation
Boom assembly
Cable
Circumaural
Cord
Cordage
Dielectric withstanding voltage
Earcup
Earphone
Electrical
Electromagnetic
Frequency response

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Harmonic
Hertz
Impedance
Insulation
Linearity
Magnetic field
Nuclear
Psychoacoustic
Staycord
Sea level
Voice coil

6.9 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity.
DLA - CC

(Project 5965-2014-001)

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